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ASSIGNMENT - 2

Microservices and Event-Driven Architecture in an E-commerce Platform

- A hypothetical e-commerce platform, shopswift is designed using microservices (MSA) to enhance scalability and maintainability. Each core business domain - user management, product catalog, shopping cart, order processing, payment :- It is developed as an independent service. AN Event-Driven Architecture is implement using an event broker to decouple services and facilitate asynchronous communication.

Application of MSA + EDA

- Order placed - When a user placed orders, the order service emits an order placed event.

- Stock updated - The inventory service listen for order placed, updates stock + emit stock updates

- Payment complete - The payment service processes the payment.

- Notification service is used to multiple events to send real-time-updates.

SOLID Principles

- Single Responsibility Principle (SRP) - Services focus on one business capability

- Open/closed principle (OCP) - Services can be extended with new features

- Liskov Substitution principle (LSP) - Interfaces for Services like payment can be substituted with new implementation

- Interface Segregation Principle - APIs are designed with Specific endpoints for client needs.

- Dependency Inversion Principle - Services depends on abstractions rather than concrete implementation

DRX and KISS Principles.

DRY (Don't Repeat yourself)

- Reusable utility libraries for authentication and logging shared across services.
- Shared Messages schema contracts for events to avoid duplication

KISS (Keep It simple, stupid):

- Services expose only essential endpoints
- Event flows follow simple pub/sub models to reduce coupling and increasing fault tolerance